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**Superconductivity-Induced Shift in Resonant Field in a
Superconductor-Ferromagnet-Superconductor Trilayer**

**Effetto della superconduttività sul campo di risonanza in un
trilayer superconduttore-ferromagnete-superconduttore**

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Sinossi

In questa tesi vengono studiati gli effetti della superconduttività sulla dinamica della magnetizzazione in strutture multistrato superconduttore-ferromagnete-superconduttore. Dopo un'introduzione teorica al ferromagnetismo, alla superconduttività, alla risonanza ferromagnetica e ai fenomeni di prossimità, vengono presentati i metodi sperimentali adottati e i risultati ottenuti. Mediante misure di risonanza ferromagnetica condotte in un ampio intervallo di frequenze e temperature, è stata osservata una variazione del campo di risonanza al di sotto della temperatura critica, in accordo con la letteratura. L'effetto scompare in campioni privi di una delle due regioni superconduttive, confermando il ruolo essenziale del doppio accoppiamento di prossimità. Sono inoltre discussi due esperimenti aggiuntivi volti a studiare l'origine di questo fenomeno, verificandone la dipendenza dallo spessore di uno strato intermedio di rame e dalla potenza del campo di sonda.

Abstract

This thesis investigates the effects of superconductivity on the magnetization dynamics of superconductor-ferromagnet-superconductor multilayer structures. After a theoretical introduction to ferromagnetism, superconductivity, ferromagnetic resonance, and proximity effects, the experimental methods and the resulting measurements are presented. Ferromagnetic resonance measurements carried out over a wide range of frequencies and temperatures reveal a shift in the resonant field below the critical temperature, in agreement with the existing literature. This effect vanishes in samples lacking one of the two superconducting layers, confirming the essential role of double proximity coupling. Three additional experiments are discussed to investigate the origin of this phenomenon, examining how the shift depends on the thickness of an intermediate copper layer and on the microwave power.

Introduction

This thesis presents an investigation of the magnetization dynamics in superconductor–ferromagnet–superconductor thin-film multilayers, conducted during my summer internship at the Cavendish Laboratory in Cambridge under the supervision of Professor Chiara Ciccarelli. During this period, I had the opportunity to participate in several experiments conducted by her group aimed at understanding the origin of the change in magnetic anisotropy induced by the superconducting layers in these trilayer systems.

The field of superconducting spintronics studies how the spin state of magnetic materials can be manipulated utilising superconductors. These proximised systems have huge potential in fields such as quantum computation since they could provide a low-energy solution to quantum memory storage. Therefore, superconducting spintronics offers a pathway towards energy-efficient quantum computing which has not been realised yet.

An example of how a superconducting phase can influence the magnetization of a material is the shift in the resonant field reported in Refs. in [1], [2], [3]. Ferromagnetic resonance is an experimental technique used to probe the magnetization of a ferromagnetic sample subjected to a constant and an oscillating field by measuring how the absorbed power changes after tuning the amplitude of the constant field and the frequency of the oscillating field. This effect is described by the Kittel equation (1.31), that connects the values of the frequency and field at which the maximum absorption (resonance) is obtained. When the two superconducting layers are introduced it is observed that a lower external field is required to obtain the resonance condition, that is now described by the modified Kittel equation (1.66), meaning that superconductivity generates an additional magnetic field in the ferromagnetic layer.

There is currently no explanation for the microscopic mechanism that originates this effect, however there are many possible explanations, among which is the Anderson-Higgs mechanism involving magnons in the ferromagnetic layer. The goal of the experiments conducted during my internship and presented in this thesis was to test this theory by observing how the resonant

field varied upon introducing metallic spacers between the ferromagnetic and superconducting layers.

The first chapter introduces the theoretical concepts necessary to interpret the experimental results. These include the fundamental properties of ferromagnetic and superconducting materials and their proximity effects, the theory of ferromagnetic resonance, the experimental technique used to probe the samples, and a review of the existing literature reporting a superconductivity-induced shift in the resonant field, together with its proposed explanations.

The second chapter provides a description of all the samples used in the study along with the experimental setup used to perform the measurements.

Finally, the third chapter presents the data collected during the experiments and discusses how they relate to the results reported in the literature and to the theoretical model under investigation.

Chapter 1

Magnetism and Superconductivity

This chapter introduces the theoretical tools necessary to understand both the experiments performed in this study and their results. Section 1.1 covers the main properties of ferromagnetic materials and ferromagnetic resonance, which is the experimental technique used to probe the magnetization of the samples. For this section the main references are Refs. [4], [5]. Section 1.2 presents the main phenomenological properties of superconducting materials along with the theories used to describe them (Ginzburg-Landau theory and the BCS theory), for this section the main reference is Ref. [6]. Finally, section 1.3 covers the proximity effects in superconductor-ferromagnet interfaces (Ref. [7]) and the experimental results known in literature (Refs. [1], [2], [3]) about superconductor-ferromagnet-superconductor trilayers, which are the object of this study.

1.1 Ferromagnetism

1.1.1 Maxwell's equations in matter

The evolution of the electric field $\mathbf{E}$ and the magnetic flux density $\mathbf{B}$ is governed by Maxwell's equations:

$$\left\{ \begin{array}{l} \nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0} \\ \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \cdot \mathbf{B} = 0 \\ \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \end{array} \right. \quad \begin{array}{l} (1.1) \\ (1.2) \\ (1.3) \\ (1.4) \end{array}$$

where ρ is the electric charge density, $\mathbf{J}$ is the electric current density. These equations, together with the Lorentz force law,

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (1.5)$$

which gives the force acting on a particle of charge q moving with velocity $\mathbf{v}$ in an electromagnetic field $(\mathbf{E}, \mathbf{B})$, constitute the theory of classical electrodynamics.

From Eq. 1.3, $\mathbf{B}$ can be written as the curl of a vector field $\mathbf{A}$ called the vector potential

$$\mathbf{B} = \nabla \times \mathbf{A}. \quad (1.6)$$

Substituting Eq. (1.6) in Eq. (1.2) we get

$$\nabla \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} \right) = \mathbf{0} \quad (1.7)$$

which implies that $\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t}$ can be written as the negative gradient of a scalar field V , called the scalar potential

$$\begin{aligned} \mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} &= -\nabla V \\ \mathbf{E} &= -\nabla V - \frac{\partial \mathbf{A}}{\partial t} \end{aligned} \quad (1.8)$$

The energy density of the electromagnetic field is given by

$$u = \frac{\varepsilon_0}{2} E^2 + \frac{1}{2\mu_0} B^2. \quad (1.9)$$

The derivation can be found in Ref. [4]. Therefore, the total energy required to generate an electromagnetic field $(\mathbf{E}, \mathbf{B})$ from a configuration with no field is

$$W = \int u d^3\mathbf{r} = \int \left(\frac{\varepsilon_0}{2} E^2 + \frac{1}{2\mu_0} B^2 \right) d^3\mathbf{r} \quad (1.10)$$

When dealing with matter, it is useful to introduce the following macroscopic quantities:

- The polarization vector, defined as the electric dipole moment per unit volume:

$$\mathbf{P} = \frac{d\mathbf{p}}{dV}. \quad (1.11)$$

- The magnetization vector, defined as the magnetic dipole moment per unit volume:

$$\mathbf{M} = \frac{d\mathbf{m}}{dV}. \quad (1.12)$$

- The electric displacement field:

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P}. \quad (1.13)$$

- The magnetic field:

$$\mathbf{H} = \frac{\mathbf{B}}{\mu_0} - \mathbf{M}. \quad (1.14)$$

In the presence of polarization and magnetization, the charge and current densities can be expressed as

$$\rho = \rho_f + \rho_b \quad (1.15)$$

where $\rho_b = \nabla \cdot \mathbf{P}$ is the bound charge density due to polarization and ρ_f is the free charge density due to macroscopic charges, and

$$\mathbf{J} = \mathbf{J}_f + \mathbf{J}_b + \mathbf{J}_p \quad (1.16)$$

where $\mathbf{J}_b = \nabla \times \mathbf{M}$ is the bound current density due to magnetization, $\mathbf{J}_p = \frac{\partial \mathbf{P}}{\partial t}$ is the polarization current, and $\mathbf{J}_f$ is the free current density due to macroscopic currents.

Using these definitions, Maxwell's equations can be rewritten in the equivalent form:

$$\left\{ \begin{array}{l} \nabla \cdot \mathbf{D} = \rho_f \\ \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \cdot \mathbf{B} = 0 \\ \nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t} \end{array} \right. \quad (1.17)$$

$$(1.18)$$

$$(1.19)$$

$$(1.20)$$

The electric and magnetic susceptibilities are defined as:

$$\chi_e = \frac{1}{\varepsilon_0} \frac{\partial \mathbf{P}}{\partial \mathbf{E}} \quad (1.21)$$

$$\chi_m = \frac{\partial \mathbf{M}}{\partial \mathbf{H}} \quad (1.22)$$

Since $\frac{\partial \mathbf{A}}{\partial \mathbf{B}}$ is a tensor such that

$$\left(\frac{\partial \mathbf{A}}{\partial \mathbf{B}} \right)_{ij} = \frac{\partial A_i}{\partial B_j},$$

χ_e and χ_m are in general tensors depending on position $\mathbf{r}$. If a material is homogeneous, they do not depend on $\mathbf{r}$; if it is isotropic, they reduce to scalar quantities. Materials with $\chi_m > 0$ are called paramagnets, while those with $\chi_m < 0$ are called diamagnets. Materials in which χ_e and χ_m do not depend on the fields are called linear.

1.1.2 Properties of ferromagnetic materials

Ferromagnets are nonlinear magnetic materials that, unlike paramagnets, do not require external fields to sustain their magnetization. This behavior is caused by several interactions, the most important being the exchange interaction, which forces the electrons in the material to align their magnetic dipoles. In contrast, linear magnetic materials can often be treated as collections of non-interacting dipoles. The typical response of the magnetization $\mathbf{M}$ of a ferromagnetic material as a function of the external magnetic field $\mathbf{H}$ is shown in Fig. 1.1.

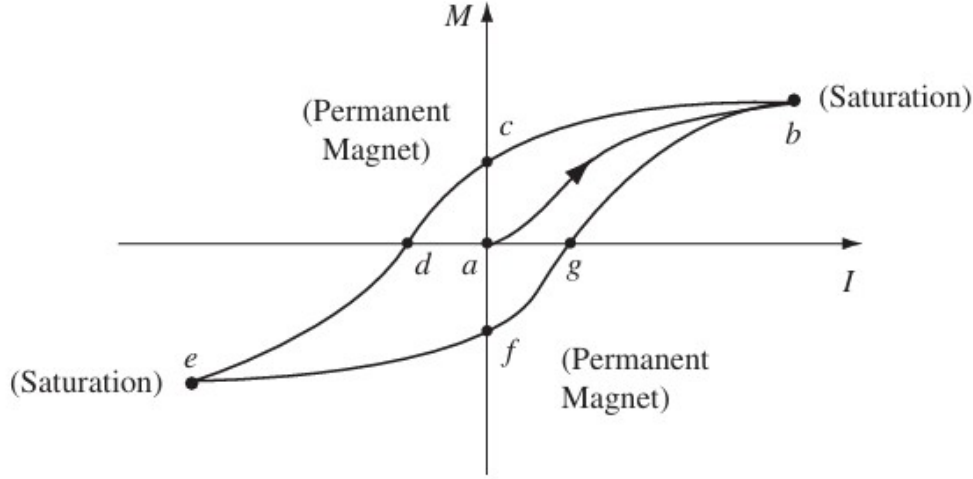


Figure 1.1: Magnetization M of a ferromagnetic material as a function of the current I flowing through a coil wrapped around the sample, which is proportional to the external magnetic field H reproduced for Ref. [4].

As H is increased, the magnetization grows until it reaches its saturation value, corresponding to the state in which all magnetic dipoles are aligned. When the magnetic field is then reduced back to zero, the magnetization does not retrace the same path; instead, it retains a finite value known as the residual magnetization. If the magnetic field is increased and decreased in the opposite direction, the magnetization again saturates, eventually reaching the same magnitude with opposite sign. The closed path traced by the magnetization during this cycle is called the hysteresis loop.

This property, namely that the magnetization depends not only on the applied field but also on the history of the material, makes ferromagnets particularly useful for the development of magnetic memory devices.

As explained earlier, the behavior of ferromagnetic materials arises from different contributions to their total energy. The most important of these is the exchange interaction, a purely quantum-mechanical effect originating from the fact that electrons are indistinguishable fermions and therefore must have an antisymmetric total wavefunction. The combination of electrostatic interactions and antisymmetric wavefunctions can be described by the effective Hamiltonian

$$H_{ex} = \sum_{i,j} -J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j \quad (1.23)$$

where J_{ij} is the exchange constant and determines whether the interaction

favors parallel ($J_{i,j} > 0$) or antiparallel ($J_{i,j} < 0$) spin alignment. For ferromagnetic materials $J_{i,j} > 0$ and triplets states with total spin $s = 1$ are favored:

$$|1, 1\rangle = |\uparrow\uparrow\rangle$$

$$|1, 0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$$

$$|1, -1\rangle = |\downarrow\downarrow\rangle$$

Where, in the state $|s, m\rangle$, s is the quantum number corresponding to $\mathbf{S}^2 = (\mathbf{S}_1^2 + \mathbf{S}_2^2)^2$ and m is the quantum number corresponding to $\mathbf{S}_z = \mathbf{S}_{1,z} + \mathbf{S}_{2,z}$. As explained in Ref. [5] the total exchange energy of a ferromagnetic material, in the continuous approximation, can be written as:

$$E_{ex} = \frac{2JS^2z}{a} \int (\nabla m_z)^2 dV \quad (1.24)$$

where a and z are constants that depend on the crystal structure of the material and $\mathbf{m} = \frac{\mathbf{M}}{M_s}$.

The Zeeman energy arises from the interaction between the magnetic moments and the external magnetic field and it is given by the equation:

$$E_z = -\mu_0 \int \mathbf{M} \cdot \mathbf{H} dV \quad (1.25)$$

Because $\nabla \cdot \mathbf{H} = -\nabla \cdot \mathbf{M}$ (as can be derived from Eq. 1.14) then whenever $\mathbf{M}$ changes without continuity, at the edges of a sample for example, the magnetic field diverges and this produces demagnetizing fields which fill space and which cost $\frac{B^2}{2\mu_0}$ Joules of energy per cubic metre. The energy associated with the demagnetizing field is called demagnetization energy and it takes the value:

$$-\frac{\mu_0}{2} \int \mathbf{M} \cdot \mathbf{H}_d dV \quad (1.26)$$

where $\mathbf{H}_d$ is the demagnetizing field and the integral is taken over the volume of the sample. This interaction is responsible for the formation of magnetic domains in ferromagnetic materials. Finally, the anisotropy energy, arising from the spin-orbit interaction and the partial quenching of the orbital angular momentum, together with the shape anisotropy, which is the demagnetizing energy associated with the geometry of the sample, gives rise to magnetic anisotropy, meaning that ferromagnetic materials possess a magnetic easy axis, along which they are easy to magnetize, and a hard axis, along which they are harder to magnetize.

1.1.3 Ferromagnetic resonance

A system with angular momentum $\mathbf{L}$ and magnetic moment $\boldsymbol{\mu} = \gamma\mathbf{L}$ subject to an external magnetic field $\mathbf{H}$ evolves according to the following equation

$$\frac{d\mathbf{L}}{dt} = \boldsymbol{\mu} \times \mathbf{H}. \quad (1.27)$$

The magnetization of a ferromagnetic sample is proportional to the sum of the magnetic moments of its electrons. Its dynamics is therefore expected to follow the equation

$$\frac{d\mathbf{M}}{dt} = \gamma\mathbf{M} \times \mathbf{H} \quad (1.28)$$

where $\mathbf{H}$ is the external applied magnetic field and γ is the gyromagnetic ratio of an electron. However this form predicts perpetual precession without relaxation while experimentally, the magnetization is observed to align with the applied field after a finite time. To account for this, damping is introduced through the Gilbert term, which models a viscous-like relaxation. The full dynamics is described by the Landau–Lifshitz–Gilbert (LLG) equation

$$\frac{1}{\gamma} \frac{d\mathbf{M}}{dt} = \mathbf{M} \times \mathbf{H} + \frac{\alpha}{M_s} \left(\mathbf{M} \times \frac{d\mathbf{M}}{dt} \right) \quad (1.29)$$

where α is the dimensionless Gilbert damping parameter and M_s is the saturation magnetization.

Ferromagnetic resonance (FMR) is observed when a ferromagnetic material is subjected to both a constant (DC) magnetic field and a perpendicular oscillating (AC) magnetic field, typically at microwave (MW) frequencies. By linearizing and solving the Landau–Lifshitz–Gilbert (LLG) equation under these conditions, it can be shown that the power absorption profile of the system follows a Lorentzian line shape

$$P_{abs} = \frac{A}{(H - H_{res})^2 + B}$$

which is characteristic of resonance phenomena. The maximum absorbed power is obtained when $H = H_{res}$

When an external magnetic field $\mathbf{H}$ is applied, the electronic energy levels split due to the Zeeman effect. Resonance occurs when the frequency of the transverse oscillating field matches the energy difference between Zeeman levels

$$\Delta E = g\mu_B B = h\nu \quad (1.30)$$

From the LLG equation, one can derive the relation between the resonance frequency ω_{res} and the resonance field H_{res} , known as the Kittel equation. Its form depends on the sample geometry and the orientation of the static field. For a thin film with an in-plane DC field, it reads

$$\omega_{res} = \gamma \sqrt{H_{res}(H_{res} + M_{eff})} \quad (1.31)$$

The Gilbert damping can also be determined by measuring the linewidth of the resonance peak as a function of frequency via the relation

$$\Delta H = \Delta H_0 + \frac{2\pi f \alpha}{\mu_0 \gamma} \quad (1.32)$$

1.2 Superconductivity

1.2.1 Properties of Superconducting Materials

The conductivity of a material is defined by the constitutive relation:

$$\mathbf{J} = \sigma \mathbf{E} \quad (1.33)$$

Here $\mathbf{J}$ and $\mathbf{E}$ are, respectively, the electrical current density and the electric field at some point in the material.

The resistivity $\rho = \sigma^{-1}$ of most metals decreases as the temperature is lowered, eventually reaching a finite value determined by the presence of impurities. However there are some materials where something different happens: when a superconducting material is cooled below a certain critical temperature T_c , its resistivity drops to zero. This is an example of a phase transition, which is a physical process where the properties of a macroscopic system change abruptly upon variation of a thermodynamic variable. Fig. 1.2 shows the difference in the temperature dependence of resistivity between a superconductor and a normal metal.

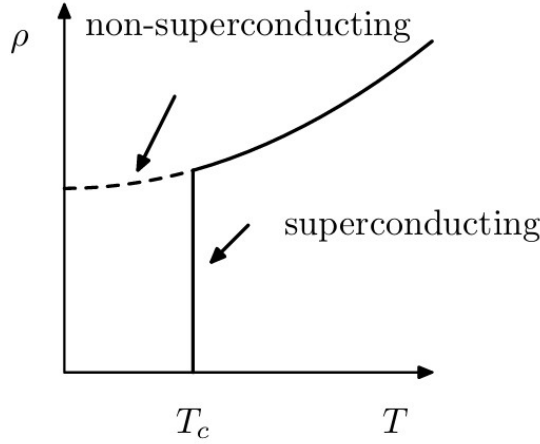


Figure 1.2: Temperature dependence of the resistivity for a normal metal and a superconducting metal reproduced for Ref. [6].

Even though superconductors were historically discovered through the observation of zero resistivity, today this property is not regarded as the true fingerprint of superconductivity. The fundamental proof that a material is superconducting is the demonstration of the Meissner-Ochsenfeld effect, namely the complete expulsion of a weak external magnetic field. This is because the magnetic behavior of superconductors below their critical temperature cannot be fully explained by zero resistivity alone.

As an example, one can consider a superconducting sample in the four configurations shown in Fig. 1.3.

Starting from configuration (I), if the sample is cooled below T_c while the external field is kept at zero, configuration (II) is obtained. If the external field is increased, the fact that $\sigma = 0$ combined with the finite value of $\mathbf{J}$ in Eq. (1.33) implies that $\mathbf{E} = \mathbf{0}$. Combining this with Maxwell's equation (1.2), it follows that $\frac{\partial \mathbf{B}}{\partial t} = 0$ which brings to configuration (IV). If instead the magnetic field is increased while the sample is kept above T_c as in (III) and then it is cooled below T_c while maintaining the external field, one again obtains $\mathbf{B} = \mathbf{0}$ inside the superconductor, showing that the magnetic field is completely expelled. This behavior cannot be explained solely in terms of zero resistance.

The complete expulsion of the magnetic field means that, for weak magnetic fields, superconductors behave as perfect diamagnets. In fact, since $\mathbf{B} = \mathbf{0}$, from Eq. (1.14) it follows that

$$\mathbf{M} = -\mathbf{H}, \quad (1.34)$$

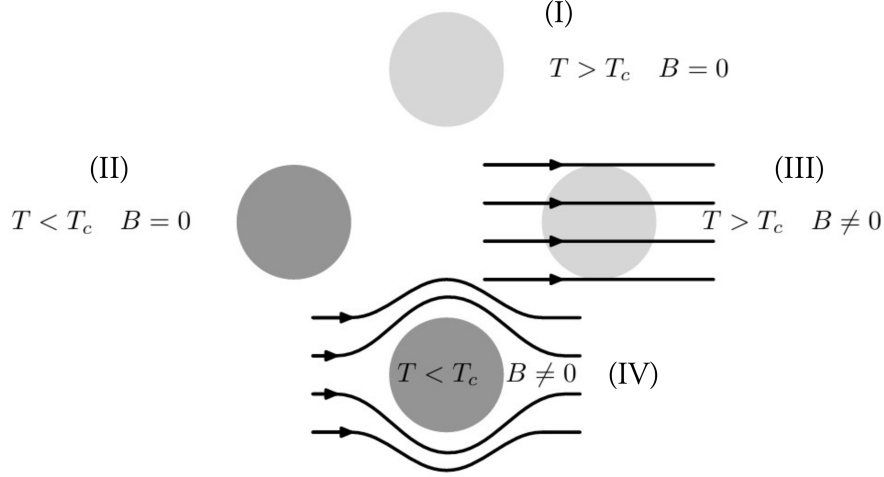


Figure 1.3: Meissner-Ochsenfeld effect in a superconductor reproduced from Ref. [6].

and combining Eq. (1.34) with Eq. (1.22), one finds

$$\chi = \frac{\partial \mathbf{M}}{\partial \mathbf{H}} = -1. \quad (1.35)$$

Superconducting materials can be divided into two categories based on their response to an increasing external magnetic field below T_c . In type I superconductors, the magnetic field $\mathbf{B}$ remains zero inside the material until the superconducting phase is destroyed at the critical field H_c . Type II superconductors, instead, exhibit two distinct critical fields: H_{c1} , the lower critical field, and H_{c2} , the upper critical field. For external fields below H_{c1} , the magnetic flux is expelled as in type I superconductors (the Meissner state). For fields above H_{c2} , superconductivity is destroyed. However, for fields between H_{c1} and H_{c2} , magnetic flux penetrates the material, and the amount of flux increases as the field is raised. This intermediate regime is known as the mixed or Abrikosov state. The behaviour of the magnetization for type I and type II superconductors is sketched in Fig. 1.4.

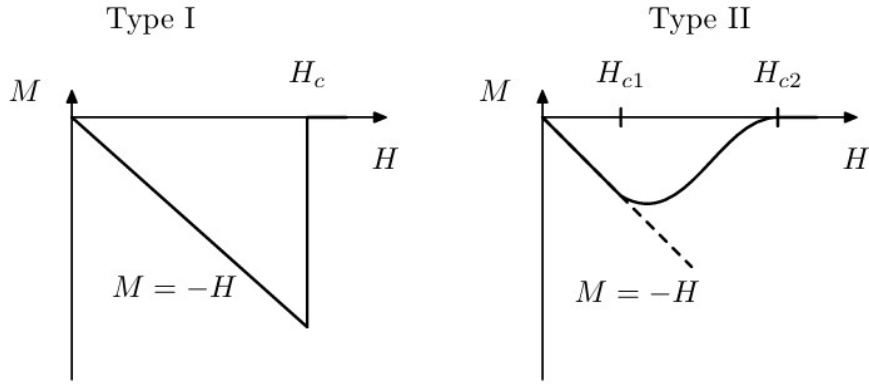


Figure 1.4: The magnetization as a function of the magnetic field in type I and type II superconductors reproduced from Ref. [6].

The critical fields vary as a function of the temperature, and they all approach zero at the critical temperature T_c . The typical phase diagrams of type I and type II superconductors, as a function of H and T are shown in Fig. 1.5.

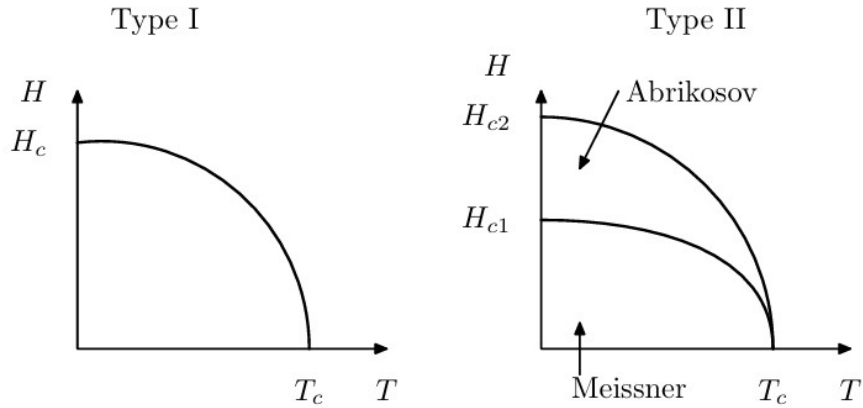


Figure 1.5: The H - T phase diagram of type I and type II superconductors reproduced from Ref. [6]. The diagrams are not to scale.

1.2.2 The London Equations

The behaviour of the current density $\mathbf{J}$ in a superconducting material is described by the London equations:

$$\frac{\partial \mathbf{J}}{\partial t} = \frac{\mathbf{E}}{\mu_0 \lambda^2} \quad (1.36)$$

$$\nabla \times \mathbf{J} = -\frac{\mathbf{B}}{\mu_0 \lambda^2} \quad (1.37)$$

These equations characterize the electrodynamic response of superconductors. The parameter λ , known as the London penetration depth, measures how deeply a magnetic field can penetrate into the superconducting material. Using Maxwell's equations, one can show that the London equations are equivalent to:

$$\mathbf{J} = -\frac{\mathbf{A}}{\mu_0 \lambda^2} \quad (1.38)$$

where the vector potential $\mathbf{A}$ plays the role of the electric field $\mathbf{E}$ in the constitutive relation (1.33).

London equations can be used to explain the Meissner-Ochsenfeld effect. By substituting Maxwell's equation (1.4) into Eq. (1.38), the following equation is obtained

$$\nabla \times (\nabla \times \mathbf{B}) = -\frac{\mathbf{B}}{\lambda^2}$$

using the vector identity

$$\nabla \times (\nabla \times \mathbf{B}) = \nabla(\nabla \cdot \mathbf{B}) - \nabla^2 \mathbf{B} \quad (1.39)$$

and using Maxwell's equation (1.3), the following equation is obtained

$$\nabla^2 \mathbf{B} = \frac{\mathbf{B}}{\lambda^2} \quad (1.40)$$

By solving this equation one can see that in the pure superconducting state the only field allowed is exponentially damped going in from an external surface. Consider for example a semi-infinite superconductor occupying the space on the positive side of the x axis with a uniform field as in Fig. 1.6.

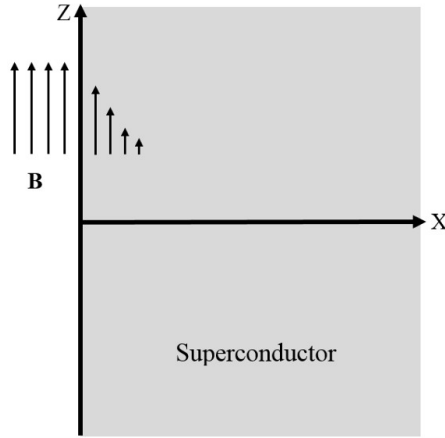


Figure 1.6: Magnetic field on the surface of a superconductor reproduced from Ref. [6].

With this geometry, the previous equation reduces to the form:

$$\frac{\partial^2 B_z}{\partial x^2} = \frac{B_z}{\lambda^2}$$

The only solution to this differential equation that is consistent with the boundary condition $B_z(x=0) = B_0$ and with finite energy (calculated from Eq. (1.10)) is

$$B_z(x) = B_0 e^{-\frac{x}{\lambda}}. \quad (1.41)$$

This exponential decay of the magnetic field near the surface of a superconductor in the Meissner state is illustrated in Fig. 1.7.

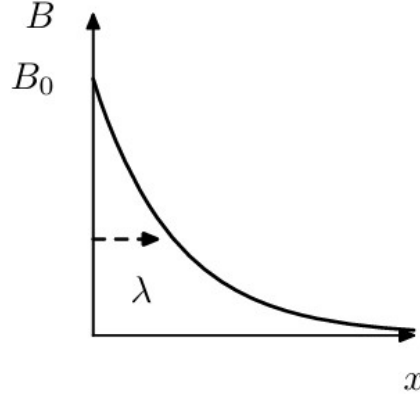


Figure 1.7: Exponential dumping of the magnetic field in a superconductor reproduced from Ref. [6].

1.2.3 The Ginzburg-Landau model

The Ginzburg–Landau model is a phenomenological theory that describes the superconducting phase transition from a thermodynamic perspective, and it is particularly useful to describe how the superconducting phase responds when brought into contact with another material. Unlike the BCS theory, which, as explained in the next section, applies only to superconductors where Cooper pairs are formed through electron–phonon interactions, the Ginzburg–Landau theory is valid for all known superconductors, regardless of the microscopic pairing mechanism. The central assumption of the theory is that a superconducting material can be described by a complex order parameter ψ such that

$$\psi = \begin{cases} 0, & T > T_c, \\ \psi(T) \neq 0, & T < T_c. \end{cases} \quad (1.42)$$

Furthermore, it assumes that the Helmholtz free energy of the superconductor depends on ψ through the relation:

$$f_s(T) \approx f_n(T) + a(T)|\psi|^2 + \frac{1}{2}b(T)|\psi|^4 \quad (1.43)$$

Here $f_s(T)$ and $f_n(T)$ are the superconducting-state and normal-state free energy densities, respectively. Depending on the sign of the parameter $a(T)$, as shown in Fig. 1.8, there are two possible behaviours for the minimum

of f_s . For $a(T) > 0$, the curve has a single minimum at $\psi = 0$, while for $a(T) < 0$ it has minima at all values satisfying $|\psi|^2 = -\frac{a(T)}{b(T)}$. The critical temperature is therefore the temperature at which $a(T) = 0$.

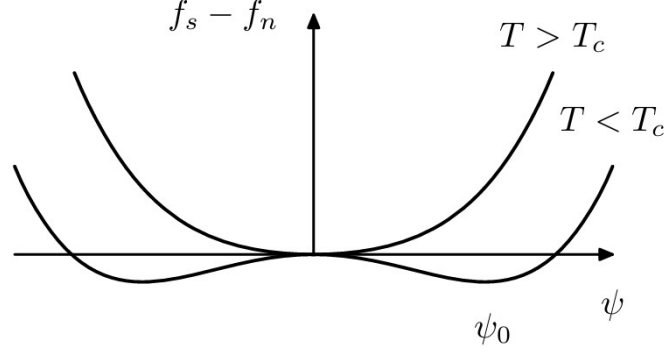


Figure 1.8: Free energy difference between the normal and superconducting states (per unit volume) as a function of the order parameter ψ reproduced from Ref. [6].

In the general case, the order parameter is not uniform in space and depends on the position. In this case, a new term depending on the gradient of $\psi(\mathbf{r})$ has to be added to Eq. (1.8). With this term free energy density becomes:

$$f_s(T) = f_n(T) + \frac{\hbar^2}{2m^*} |\nabla \psi(\mathbf{r})|^2 + a(T) |\psi(\mathbf{r})|^2 + \frac{b(T)}{2} |\psi(\mathbf{r})|^4 \quad (1.44)$$

The new parameter m^* has dimensions of mass, and it plays the role of an effective mass for the quantum system with macroscopic wave function $\psi(\mathbf{r})$.

In order to find the order parameter at the equilibrium, one must minimize the total free energy of the system:

$$F_s(T) = F_n(T) + \int \left(\frac{\hbar^2}{2m^*} |\nabla \psi(\mathbf{r})|^2 + a(T) |\psi(\mathbf{r})|^2 + \frac{b(T)}{2} |\psi(\mathbf{r})|^4 \right) d^3\mathbf{r} \quad (1.45)$$

To find the minimum, the infinitesimal change in the total free energy δF must be equal to zero for any change in the order parameter $\delta\psi$. This leads to the following equation:

$$-\frac{\hbar^2}{2m^*} \nabla^2 \psi + a\psi + b\psi|\psi|^2 = 0 \quad (1.46)$$

This is a Schrödinger-like equation that allows to find the order parameter $\psi(\mathbf{r})$ in equilibrium conditions. However, unlike the usual Schrödinger equation, this is a non-linear equation because of the second term in the bracket.

The behaviour of the order parameter at the interface with a normal metal can be studied by applying Eq. (1.46) to simple system consisting of an interface between a normal metal and a superconductor with the interface lying in the yz plane separating the normal metal in the $x < 0$ region from the superconductor in the $x > 0$ region. In this case Eq. (1.46) takes the form:

$$-\frac{\hbar^2}{2m^*} \frac{d^2\psi}{dx^2} + a\psi + b\psi|\psi|^2 = 0 \quad (1.47)$$

in the region $x > 0$ with the boundary condition $\psi(0) = 0$. This equation can be solved directly and its solution is

$$\psi(x) = \psi_0 \tanh\left(\frac{x}{\sqrt{2}\xi(T)}\right) \quad (1.48)$$

as shown in Fig. 1.9.

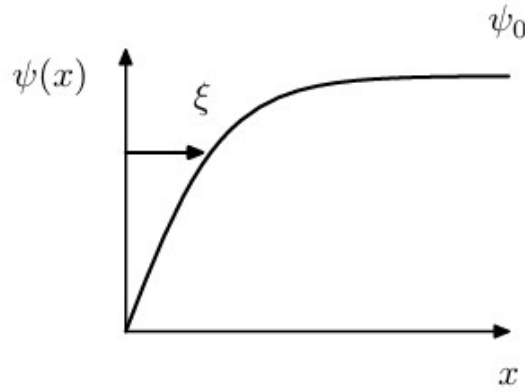


Figure 1.9: Order parameter of a superconductor near a surface reproduced from Ref. [6]. It recovers to its bulk value ψ_0 over a length scale of the coherence length, ξ .

Here ψ_0 is the value of the order parameter in the bulk far from the surface and the parameter $\xi(T)$ is defined by

$$\xi(T) = \frac{\hbar}{\sqrt{2m^*|a(T)|}} \quad (1.49)$$

This quantity, which has dimensions of length, is called the Ginzburg-Landau coherence length. It is a measure of the distance required to the order parameter to recover to its bulk value.

In the presence of a magnetic field the expression (1.44) must be extended according to the minimal substitution:

$$-i\hbar\nabla \rightarrow -i\hbar\nabla - q\mathbf{A} \quad (1.50)$$

For all known superconductors it turns out that the appropriate charge q is $-2e$ and the reason for this becomes clear in the BCS theory. With this replacement the Ginzburg-Landau free energy density of the superconductor becomes

$$f_s(T) = f_n(T) + \frac{1}{2m^*}|(-i\hbar\nabla + 2e\mathbf{A})\psi|^2 + a|\psi|^2 + \frac{b}{2}|\psi(\mathbf{r})|^4 \quad (1.51)$$

The London equation (1.38) can be shown to follow from the Ginzburg-Landau model. This can be done using the following expression for the current density:

$$\mathbf{J} = -\frac{\partial F}{\partial \mathbf{A}} \quad (1.52)$$

where is the functional derivative, which is defined by the relation:

$$\delta F = \int \frac{\partial F}{\partial \mathbf{A}} \cdot \delta \mathbf{A} d^3\mathbf{r} \quad (1.53)$$

This equation gives:

$$\mathbf{J} = -\frac{\partial F}{\partial \mathbf{A}} = -\frac{e\hbar i}{m^*}(\psi^*\nabla\psi - \psi\nabla\psi^*) - \frac{4e^2}{m^*}|\psi|^2\mathbf{A} \quad (1.54)$$

in the bulk of the material the order parameter can be considered uniform, therefore $\nabla\psi = \nabla\psi^* = \mathbf{0}$ and one can write

$$\mathbf{J} = -\frac{4e^2|\psi|^2}{m^*}\mathbf{A} \quad (1.55)$$

which is the London equation 1.38. Comparing this with the known form, it follows that:

$$m^* = 2m_e \quad (1.56)$$

and

$$|\psi|^2 = \frac{n_s}{2} \quad (1.57)$$

where n_s is the electron density. This gives a hint for the physical meaning of the order parameter ψ and the effective mass m^* : ψ is the coherent wave function of the Cooper pairs and m^* is the mass of a single Cooper pair.

1.2.4 The BCS theory

The Ginzburg–Landau model makes it clear that the superconducting phase must originate from a coupling mechanism between electrons, as the phenomenological parameters take the values $q = -2e$ and $m^* = 2m_e$. However, the model itself does not explain the nature of this coupling. The BCS theory accounts for superconductivity by introducing an attractive interaction between electrons mediated by the exchange of lattice phonons. The electrons involved in this attractive interaction form a bound state known as a Cooper pair. Fig. 1.10 shows the Feynman diagram corresponding to this interaction.

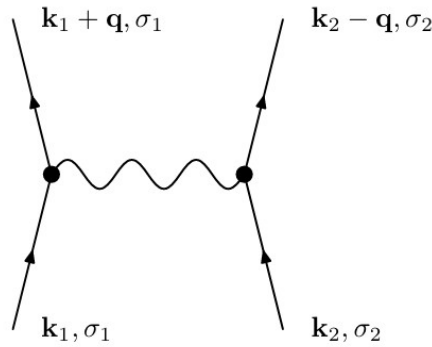


Figure 1.10: Feynmann diagram representing the interaction between electrons at the Fermi surface due to exchange of crystal lattice phonons reproduced from Ref. [6].

This effective interaction between the electrons due to exchange of phonons is of the form:

$$V_{eff}(\mathbf{q}, \omega) = \frac{|g_{\mathbf{q}}|^2}{\omega^2 - \omega_{\mathbf{q}}^2}$$

where the phonon has wave vector $\mathbf{q}$ and frequency ω . The parameter $g_{\mathbf{q}}$ is related to the matrix element for scattering an electron from state $\mathbf{k}$ to $\mathbf{k} + \mathbf{q}$ as shown in Fig. 1.10 and is of the order

$$g_{\mathbf{q}} \sim \sqrt{\frac{m}{M}}$$

where m is the effective mass of the electrons at the Fermi surface and M is the mass of the lattice ions. Since $\frac{m}{M}$ is of order 10^{-4} , typically, the electrons and phonons are only weakly coupled, this entails that all the diagrams of higher

order than the one in Fig. 1.10 (that involve more vertices) can be neglected. By neglecting the dependence of the interaction the wave vector and replacing the interaction with an approximate one that effectively averages over all values of $\mathbf{q}$, it takes the form:

$$V_{eff}(\mathbf{q}, \omega) = \frac{|g_{eff}|^2}{\omega^2 - \omega_D^2}$$

where ω_D is the Debye frequency of phonons. This interaction only involves electrons in a thin shell of thickness $2\hbar\omega_D$ around the Fermi surface (shown in Fig. 1.11), since only these states can change energy without violating the Pauli exclusion principle.

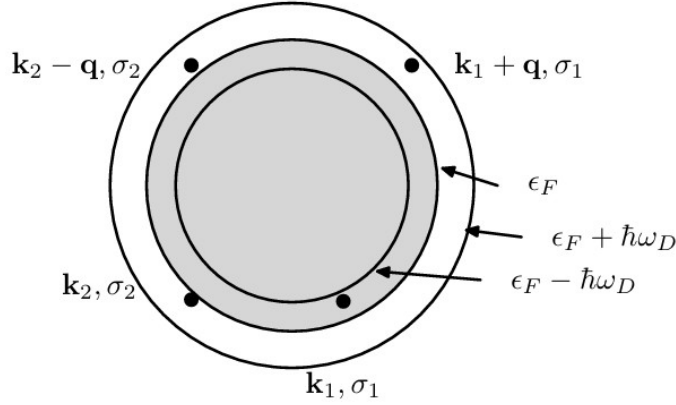


Figure 1.11: The effective electron-electron interaction near the Fermi surface reproduced from Ref. [6].

Since $\hbar\omega_D \ll K_B T$ the interaction is always attractive and it can be approximated as:

$$V_{eff} = -|g_{eff}|^2 \quad (1.58)$$

The corresponding effective Hamiltonian for the effective electron-electron interaction is therefore:

$$H = -|g_{eff}|^2 \sum c_{\mathbf{k}_1 + \mathbf{q}, \sigma_1}^\dagger c_{\mathbf{k}_2 - \mathbf{q}, \sigma_1}^\dagger c_{\mathbf{k}_1, \sigma_1} c_{\mathbf{k}_2, \sigma_2}^\dagger \quad (1.59)$$

where $c_{\mathbf{k}, \sigma}^\dagger$ and $c_{\mathbf{k}, \sigma}$ respectively denote the creation and destruction operator of an electron with momentum $\mathbf{k}$ and spin σ respectively. The sum is over all values of $\mathbf{k}_1, \sigma_1, \mathbf{k}_2, \sigma_2$ and $\mathbf{q}$ with the restriction that the electron energies involved are all within the range $\pm\hbar\omega_D$ of the Fermi surface as in Fig. 1.11.

The wave function for a pair of electrons near the Fermi surface can be written as:

$$\Psi(\mathbf{r}_1, \sigma_1, \mathbf{r}_2, \sigma_2) = e^{i\mathbf{k}_c \cdot \mathbf{R}} \varphi(\mathbf{r}_1 - \mathbf{r}_2) \phi_{\sigma_1, \sigma_2}^{spin}$$

where $\mathbf{R}$ and $\mathbf{k}_c$ are the position and momentum of the center of mass respectively. In the ground state $\mathbf{k}_c = 0$ and in most superconductors, the pair is in a singlet state:

$$\phi_{\sigma_1, \sigma_2}^{spin} = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

therefore, since electrons are fermions and their wave functions must be antisymmetrical, $\varphi(\mathbf{r}_1 - \mathbf{r}_2)$ must be an even function. Substituting the wave function in Eq. (1.60) into the Schrödinger's equation gives the following expression for the energy of a Cooper pair:

$$E = -2\hbar\omega_D e^{-\frac{1}{\lambda}} \quad (1.60)$$

with:

$$\lambda = |g_{eff}|^2 g(\varepsilon_F)$$

where $g(\varepsilon_F)$ is the density of state evaluated at the Fermi level. This confirms the existence of a bound state between two electrons at the Fermi surface. This entails that the entire Fermi surface is unstable to the creation of Cooper pairs. The wave function on the entire system is given by the coherent wave function:

$$|\Psi_{BCS}\rangle = C \exp \left(\sum_{\mathbf{k}} \alpha_{\mathbf{k}} P_{\mathbf{k}\dagger} \right) |0\rangle$$

where

$$P_{\mathbf{k}}^\dagger = c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger \quad (1.61)$$

is the operator that creates a Cooper pair and $|0\rangle$ is the vacuum state where no Cooper pair is present. Using the properties of the creation and destruction operators (the calculations are shown in Ref. [6]) the wave function can be rewritten as:

$$|\Psi_{BCS}\rangle = \prod_{\mathbf{k}} (u_{\mathbf{k}}^* + v_{\mathbf{k}}^* P_{\mathbf{k}}^\dagger) |0\rangle \quad (1.62)$$

where

$$u_{\mathbf{k}}^* = \frac{1}{1 + |\alpha_{\mathbf{k}}|^2}$$

$$v_{\mathbf{k}}^* = \frac{\alpha_{\mathbf{k}}}{1 + |\alpha_{\mathbf{k}}|^2}$$

The values of $u_{\mathbf{k}}^*$ and $v_{\mathbf{k}}^*$ can be found using the variational method and minimizing the energy functional

$$E = \langle \Psi_{\text{BCS}} | H | \Psi_{\text{BCS}} \rangle$$

This procedure gives the following system of equations:

$$\begin{cases} |u_{\mathbf{k}}|^2 = \frac{1}{2} \left(1 + \frac{\varepsilon_{\mathbf{k}} - \mu}{E_{\mathbf{k}}} \right) \\ |v_{\mathbf{k}}|^2 = \frac{1}{2} \left(1 - \frac{\varepsilon_{\mathbf{k}} - \mu}{E_{\mathbf{k}}} \right) \\ E_{\mathbf{k}} = \sqrt{(\varepsilon_{\mathbf{k}} - \mu)^2 + |\Delta|^2} \\ \Delta = |g_{eff}|^2 = \sum_{\mathbf{k}} u_{\mathbf{k}} v_{\mathbf{k}}^* \end{cases}$$

where Δ is called the BCS gap parameter. The system leads to the following expression for the gap parameter at $T = 0$:

$$|\Delta| = 2\hbar\omega_D e^{-\frac{1}{\lambda}} \quad (1.63)$$

From this, the expression for the gap parameter at finite temperature can be obtained by taking $|\Psi_{\text{BCS}}\rangle$ as the reference state and considering small excitations relative to it. The resulting formula follows from expressing the creation and destruction operators in terms of the quasiparticle creation and destruction operators and using their Fermi–Dirac occupation statistics.

1.3 Proximity Effects

1.3.1 Superconductor-Ferromagnet Interfaces

Due to their opposite ordering, ferromagnetism and superconductivity are highly incompatible: the exchange field, in a magnetically ordered state, tends to align the spins of the Cooper pairs in the same direction, thus preventing a pairing effect. This is called paramagnetic effect and, because of it, the coexistence of singlet superconductivity with ferromagnetism is very unlikely in bulk materials.

However, at the interface between a ferromagnetic and a superconducting material, the Cooper pairs can penetrate the ferromagnet at some distance and induce superconducting-like properties (this phenomenon is called proximity effect), at the same time the leakage of the Cooper pairs weakens the superconductivity near the interface (this effect is called the inverse proximity effect). These effects result in a decrease of the superconducting transition temperature in a thin superconducting layer. If the thickness of a superconducting layer is smaller than a critical one, the proximity effect totally suppresses the superconducting transition. This effect causes the coherent wave function ψ in the ferromagnet (that is associated to the Cooper pairs that crossed the interface) to decay exponentially, as in a superconductor-metal interface, but modulated by an additional oscillating factor due to the exchange field.

The oscillatory behavior of the order parameter can be understood qualitatively using the generalized Ginzburg–Landau theory, as done in Ref. [7]. The free-energy density in the ferromagnet may be written as:

$$f_s = f_n + a(H, T)|\psi|^2 + \gamma(H, T)|\nabla\psi|^2 + \frac{\eta(H, T)}{2}|\nabla^2\psi|^2 + \frac{b(H, T)}{2}|\psi|^4 \quad (1.64)$$

By minimizing f_s and choosing the x axis perpendicular to the interface (so that ψ depends only on x), the following equation for the order parameter ψ is obtained:

$$a\psi - \gamma \frac{\partial^2 \psi}{\partial x^2} + \frac{\eta}{2} \frac{\partial^4 \psi}{\partial x^4} = 0.$$

The solutions to this equation are of the form:

$$\psi = \psi_0 e^{kx}$$

with a complex wave vector k given by:

$$k = k_1 + ik_2$$

and

$$k_1^2 = \frac{|\gamma|}{2\eta} \left(\sqrt{1 + \frac{T - T_{ci}}{T_{ci} - T_{cu}}} - 1 \right)$$

$$k_1^2 = \frac{|\gamma|}{2\eta} \left(\sqrt{1 + \frac{T - T_{ci}}{T_{ci} - T_{cu}}} + 1 \right)$$

where $T_{cu}(H)$ is the critical temperature of the transition into the uniform superconducting state and $T_{ci}(H) > T_{cu}(H)$ is critical temperature of the non-uniform (FFLO-type) superconducting state. These temperatures are effective parameters describing the hypothetical superconducting state associated with the wave function ψ in the ferromagnetic region. Choosing ψ to be real leads the following expression:

$$\psi(x) = \psi_i e^{-k_1 x} \cos(k_2 x) \quad (1.65)$$

The behaviour of the order parameter ψ in the ferromagnet is sketched in Fig. 1.12.

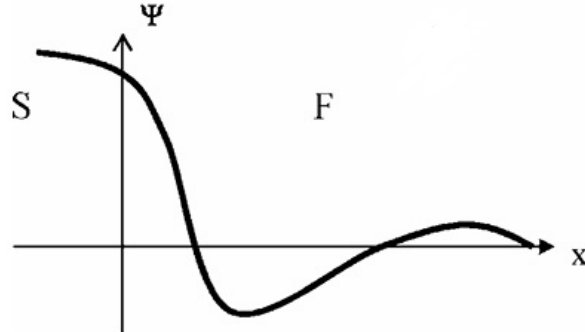


Figure 1.12: Schematic behavior of the superconducting order parameter near the superconductor-ferromagnet interface reproduced from Ref. [7].

This oscillating behaviour is a fundamental difference between the proximity effect in superconductor-ferromagnet and superconductor-metal systems, and it is at the origin of many peculiar characteristics of superconductor-ferromagnet heterostructures.

1.3.2 Shift in the resonant field

The first evidence for a shift in the FMR resonant frequency (and field) in superconductor–ferromagnet–superconductor trilayers is presented in Refs. [2], [3], and [1]. Fig. 1.13 shows a schematic representation of the trilayers used in these works, together with the static (DC) and oscillating (MW) magnetic fields employed in the FMR measurements.

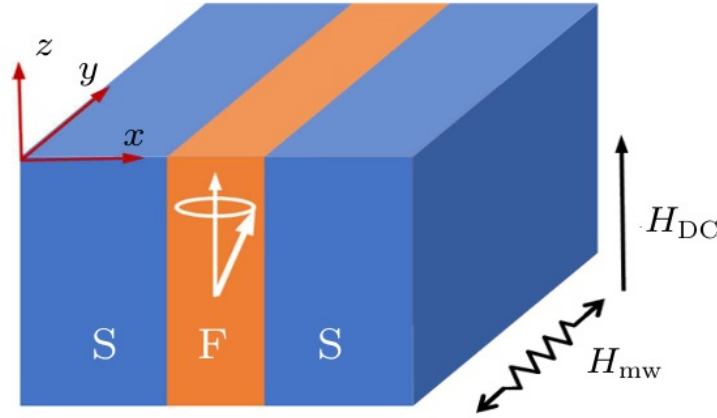


Figure 1.13: Geometry of the ferromagnetic Josephson junctions and configuration of FMR measurements from Ref. [2].

These studies show that when FMR measurements are conducted on an S/F/S thin-film trilayer, the resonant frequency and field no longer obey the standard Kittel equation 1.31. Instead, they follow the modified Kittel equation:

$$\omega_{res} = \gamma \sqrt{(H_{res} + H_a)(H_{res} + H_a + M_{eff})} \quad (1.66)$$

where H_a is an additional anisotropy field. This equation indicates that the superconducting layers generate an effective magnetic field in the same direction as the external DC field, implying that a lower external field is required to reach ferromagnetic resonance. Fig. 1.14 shows the FMR absorption spectra measured in Ref. [1] for a trilayer sample above and below the critical temperature.

In both graphs, the line corresponding to the maximum absorption (resonance) is clearly visible, and for the same frequency the corresponding resonant field is lower at $T = 1.7$ K than at $T = 9.0$ K. Fig. 1.15 shows the results of a similar experiment performed in Ref. [2] on another trilayer sample made of the same materials and grown on a SiO_2 substrate. The upper

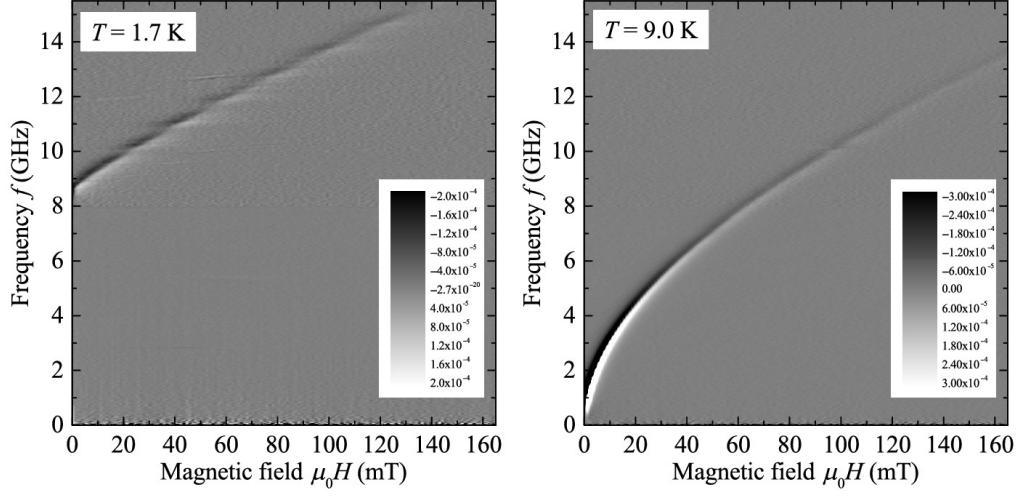


Figure 1.14: FMR absorption spectra for a sample above and below the critical temperature for a S(Nb= 110 nm)F(Py= 19 nm)S(Nb= 110 nm) trilayer sample from Ref. [1]. The grayscale is coded in absolute units.

panel shows the absorption spectra at different temperatures above and below the critical temperature of the superconducting layers. As explained in the next chapter, the experimental setup measures a voltage proportional to the derivative of the absorbed power with respect to the DC external magnetic field; therefore, the plotted data correspond to the derivative of a Lorentzian. The lower panel shows the resonant fields extracted from these data as a function of temperature, along with a schematic representation of the sample.

All the resonance lines exhibit a single Lorentz lineshape. The resonant field changes little with temperature above T_c . However, it shifts rapidly to a lower field with decreasing temperature below T_c , and the significant shift of H_r below the critical temperature demonstrates a strong proximity-induced modification of the magnetization dynamics caused by superconductivity.

Furthermore, these studies attempt to identify the mechanism responsible for this effect. More specifically, they investigate whether the additional anisotropy field is generated by supercurrents flowing through the entire Josephson junction rather than by a local proximity effect at a single S/F interface. If this hypothesis is correct, no shift should be observed in an S/F bilayer, nor in an S/I/F/S multilayer (where an insulating layer prevents Josephson coupling). Finally, an S/F/S/F multilayer should behave as a combination of an S/F/S trilayer and an S/F bilayer: above T_c the system should exhibit a single resonance line (because the two structures share

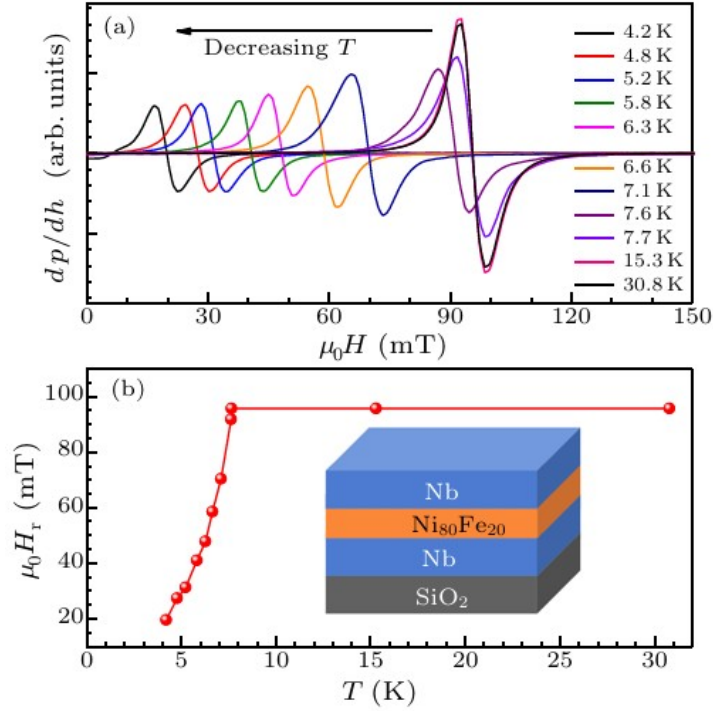


Figure 1.15: FMR spectra of a Nb(100 nm)/Ni₈₀Fe₂₀(20 nm)/Nb(100 nm) trilayer from Ref. [2]. (a) FMR spectra at various temperatures. Below the superconducting transition temperature T_c , the resonance field H_r shifts to lower values with decreasing temperature. (b) Resonance field H_r as a function of temperature. The inset shows the sample structure.

the same resonant field and frequency), whereas below T_c two distinct lines should appear (since the resonant field shifts only in the trilayer). These experiments were performed in Ref. [2], and the results are shown in Fig. 1.16.

These experiments confirmed the hypothesis: no resonant-field shift was detected in the S/F bilayer nor in the S/I/F/S multilayer, while the S/F/S/F multilayer exhibited one resonance line above T_c and two below T_c . Overall, these results demonstrate that the shift of the resonant field below the critical temperature occurs only in conducting ferromagnetic Josephson junctions.

In Ref. [1] it is shown that the dependence of the anisotropy field H_a on the temperature can be characterized by fitting it with the following power law expression:

$$H_a(T) = H_0 \left(1 - \left(\frac{T}{T_c} \right)^p \right) \quad (1.67)$$

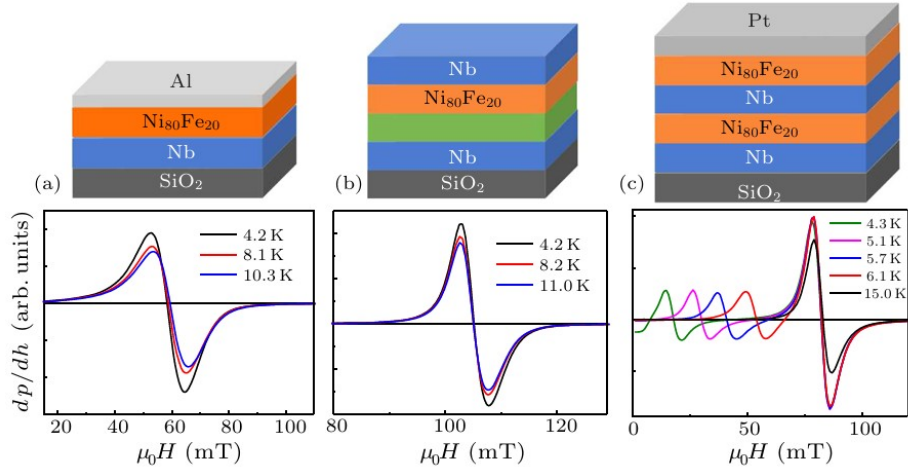


Figure 1.16: Control experiments performed in Ref. [2] on S/F bilayer, S/I/F/S, and S/F/S/F multilayers. (a) FMR spectra of a Nb(100 nm)/Ni80Fe20(20 nm) bilayer, showing no shift below T_c . (b) FMR spectra of a Nb(100 nm)/MgO(10 nm)/Ni80Fe20(20 nm)/Nb(100 nm) multilayer, again without a shift below T_c . (c) FMR spectra of a Nb(100 nm)/Ni80Fe20(20 nm)/Nb(100 nm)/Ni80Fe20(20 nm) multilayer. Above T_c there is a single resonance line, whereas below T_c two distinct lines appear: the upper NiFe layer shows no shift, while the lower NiFe layer in the S/F/S junction shifts to lower fields.

where H_0 is the effective anisotropy field at zero temperature, T_c is the critical temperature, and p is a free exponent parameter. The fit of $H_a(T)$ with equation 1.67 is shown in Fig. 1.17.

However, the exact physical mechanism responsible for this shift remained unclear. According to Ref. [2], the Meissner effect should be excluded because similar behavior would be expected in the S/I/F/S multilayer if flux focusing were the cause, as the thin MgO layer does not block magnetic flux. Since no shift is observed in such samples, the expulsion of external magnetic flux cannot explain the change in the ferromagnetic anisotropy.

A possible explanation is presented in Ref. [9]. According to this paper, the shift in the FMR resonant field is caused by the Anderson-Higgs mechanism, which is a well known mechanism in high-energy and condensed matter physics in which particles acquire mass because of the spontaneous breaking of gauge symmetry. In proximity coupled S/F/S trilayers, the spontaneous shift of the FMR frequency originates from an Anderson-Higgs-type mechanism that gives magnons a finite spectral gap and a mass. When the magnetization in the ferromagnetic layer precesses, its time-dependent

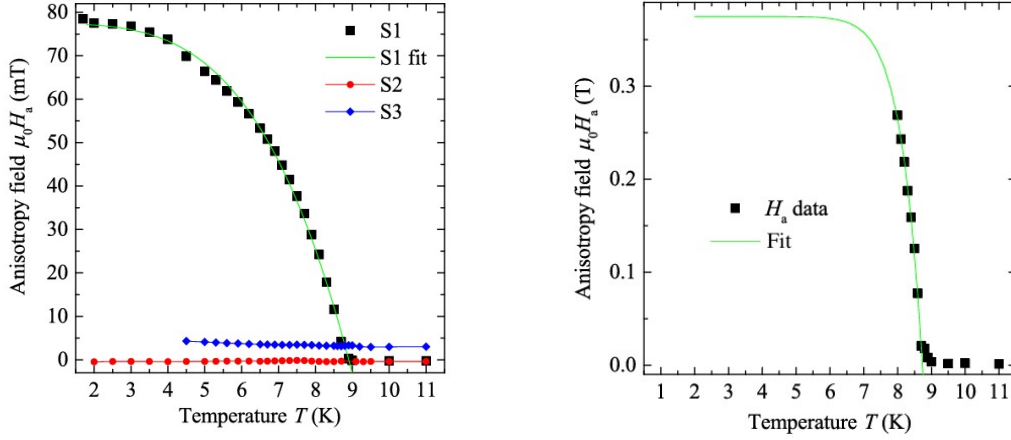


Figure 1.17: Temperature dependence of the anisotropy field measured in Ref. [1]. In the left image, black squares correspond to a symmetric S/F/S sample, red circles to an S/F/S sample with one significantly thinner superconducting layer, and blue diamonds to an S/F/I/S sample. The green curve shows the fit using equation 1.67. In the right image, black squares correspond to another symmetric S/F/S sample, and the green curve represents the corresponding fit.

magnetic flux induces curling electric fields in the adjacent superconductors, which in turn generate eddy supercurrents. As a result, magnons become massive, leading to a finite FMR frequency even at zero external magnetic field. This theory successfully predicts some features of this phenomenon such as the occurrence of the shift only in the presence of both the superconducting layers.

1.4 Summary

The aim of this investigation is to reproduce the results obtained in Refs. [2], [3], and [1], and to verify whether this theoretical interpretation holds by conducting FMR measurements on S/F/S trilayers in different conditions.

By presenting the main theoretical and phenomenological properties of superconducting and ferromagnetic materials, this chapter provided the foundation needed to interpret the experimental results discussed in the remainder of this thesis. The section on magnetization dynamics and ferromagnetic resonance explained the theory underlying the experimental technique used to obtain all the data analyzed in the final chapter. Finally, the review of

proximity effects and of the superconductivity-induced shift in the resonant field highlights the open questions that motivate this investigation.

Chapter 2

Experimental Methods and Setup

This chapter describes the details of the experimental setup used for the measurements. Section 2.1 presents the structure of the samples and the principles of the techniques used to grow and characterize them. Section 2.2 explains the experimental apparatus used to generate the magnetic field and to measure the resonant field of the samples. Finally, Section 2.3 illustrates the expected behavior of the signal measured by the setup.

In this investigation, a total of four experiments were conducted. The first two experiments aimed to reproduce the results reported in Ref. [1] by demonstrating that the shift in the resonant field follows the behavior described in literature; this was verified by performing measurements on a sample containing only one superconducting layer. The remaining two experiments were designed to test the dynamic effect theory presented in Ref. [9] by examining how the anisotropy field H_a changes under different conditions: introducing an additional metallic layer between the ferromagnetic and superconducting layers and varying the microwave power.

2.1 Sample fabrication

The samples used in this study consisted of multilayer thin films with a layer of niobium (Nb), which is a type II superconductor with a critical temperature of 9 K, a ferromagnetic layer of permalloy $\text{Ni}_{80}\text{Fe}_{20}$ (Py), followed by another niobium layer. The films were deposited on silicon (Si) substrates, which were cut into $5\text{ mm} \times 5\text{ mm}$ squares and cleaned by sonication in acetone for 2 minutes, followed by sonication in isopropanol (IPA) for 2 minutes. This cleaning process is necessary to ensure a good adhesion of the samples

to the substrate and a uniform thickness of the samples and to make the experiment reproducible. In one of the experiment two copper (Cu) layers with variable thickness were introduced between both the superconducting and the ferromagnetic layers, Fig. 2.1 shows the structure of the samples used in this experiment.

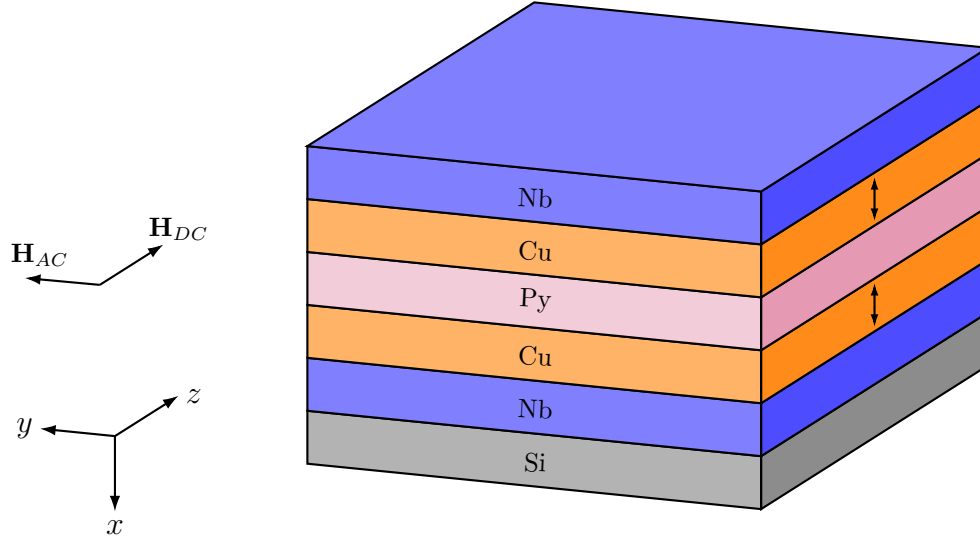


Figure 2.1: Scheme of the samples used in this study. The double arrows indicate the variable thickness of the copper layers. The figure also shows the orientation of the AC and DC magnetic fields used in the FMR measurements.

These samples were grown by DC magnetron sputtering, a physical vapour deposition (PVD) technique widely employed for the fabrication of metallic and multilayer thin films because it ensures high deposition rates and uniform coatings. Furthermore it allowed to grow the different layers in the same process, thus avoiding the exposure of intermediate layers and ensuring a good interface quality. In this method described in detail in Ref. [10]), an inert gas (argon in this case) is introduced into an ultra-high-vacuum chamber, and a high voltage is applied between the negatively charged target and the positively charged substrate. If the target material is conductive, then it is connected to direct current (DC) power supply and if the target material is insulator then it is connected to radio frequency (RF) power supply. Under the applied voltage, the gas is ionised, forming a plasma composed of positively charged ions and free electrons. The ions are accelerated by the electric field and strike the target with sufficient energy to eject atoms from it. These atoms then travel through the chamber and deposit as thin films

on the substrate positioned in their path which is kept rotating to guarantee uniform sample thickness. In magnetron sputtering the plasma is confined near the target by a magnetic field that forces electrons to follow a helical trajectory around the magnetic field lines. This increases both the sputtering yield and the overall deposition efficiency. The properties of the deposited films can be adjusted by changing sputtering parameters such as gas pressure, target–substrate distance and system temperature. Fig. 2.2 illustrates the general scheme of a sputtering device.

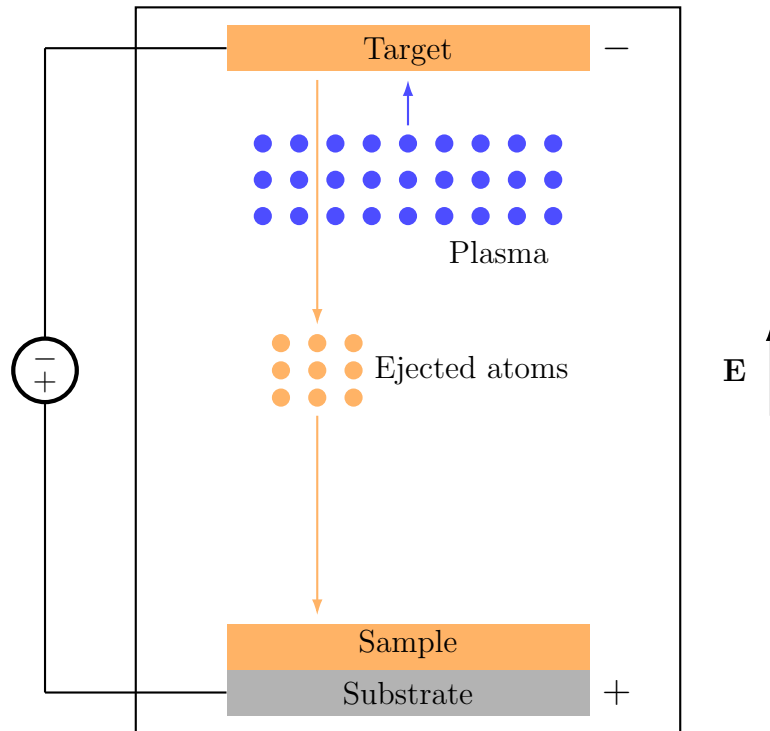


Figure 2.2: Schematic illustration of a sputtering device.

The structures investigated in this work are listed in the following tables. Table 2.1 shows the thicknesses of the layers for the samples used in the first two experiments and in the microwave-power experiments (all of which were performed on a single sample). Table 2.2 reports the thicknesses of the layers for the copper-thickness experiment.

The samples were characterized using atomic force microscopy (AFM), which is a type of scanning probe microscopy that maps the surface topography of a sample with nanometer resolution by mechanically probing it with a sharp tip. The core of an AFM consists of a cantilever with a sharp tip that

Sample	Nb thickness (nm)	Py thickness (nm)
Control Sample	80 ± 2.4	20 ± 0.6
Bilayer	$80 \pm 2.4/0$	20 ± 0.6
Microwave Power	80 ± 2.4	20 ± 0.6

Table 2.1: List of samples used in this study.

Sample	Nb thickness (nm)	Cu thickness (nm)	Py thickness (nm)
S ₁	80 ± 2.4	2 ± 0.06	20 ± 0.6
S ₃	80 ± 2.4	10 ± 0.3	20 ± 0.6
S ₅	80 ± 2.4	50 ± 1.5	20 ± 0.6
S ₅	80 ± 2.4	80 ± 1.5	20 ± 0.6
S ₆	80 ± 2.4	100 ± 3	20 ± 0.6
S ₇	80 ± 2.4	200 ± 6	20 ± 0.6

Table 2.2: List of samples used in this study.

interacts with the sample surface through intermolecular forces between the atoms at the tip and the sample that cause the cantilever to deflect. This deflection is measured by a laser beam reflected from the cantilever onto a position-sensitive photodiode. A feedback circuit maintains a constant interaction between the tip and the surface by adjusting the vertical position of the scanner. The height adjustments required to maintain the set interaction force are recorded to reconstruct the surface topography. This allowed to calculate the 3% errors on the thickness of the layers. Fig. 2.3 shows the general scheme of a setup used for AFM measurements.

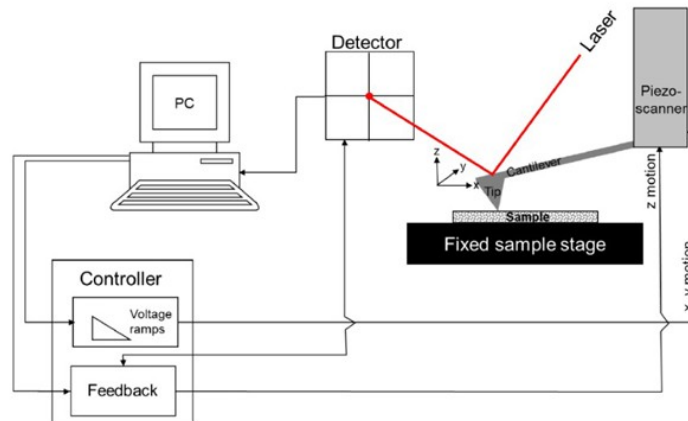


Figure 2.3: Schematic representation of an AFM setup.

2.2 Experimental Setup

The samples are mounted using vacuum grease onto a coplanar waveguide (CPW) and placed into a cryostat. A CPW is a planar transmission line structure consisting of a central conducting strip separated by narrow gaps from two ground planes, all defined on the same surface of a dielectric substrate. The signal propagates along the central conductor, while the two adjacent ground planes provide a return path. The sample is placed upside down (with the upper superconducting layer of Fig. 2.1 in contact with the CPW) because the magnetic field generated by the current in the CPW is stronger in the small region close to the surface and, due to the substrate being much thicker than the other layers, placing it the other way would make the magnetic field in the Py layer too weak to be measured. The external magnetic field and cryogenic environment are provided by the Oxford Instruments Teslatron cryostat, which contains a superconducting vector magnet with 6 T in the z-direction, 2 T in the x- and y-directions, and a base temperature of 1.5K. The sample is mounted on a sample probe, where the thickness lies in the x-direction, and is designed so that the sample is positioned between the superconducting coils to maximise the uniformity of the field. Steel coaxial cables run down the length of the sample probe and act as microwave transmission lines to the CPW. Connected to the top of the cryostat, there are copper coaxial cables. The first copper cable acts as the input and is connected to a microwave generator, which drives the AC field that excites the sample. The second cable is connected to a microwave detector, which is directly connected to a lock-in amplifier. Fig. 2.4 shows a schematic diagram of the cryostat and probe assembly while Fig. 2.5 shows an actual picture of the probe and sample.

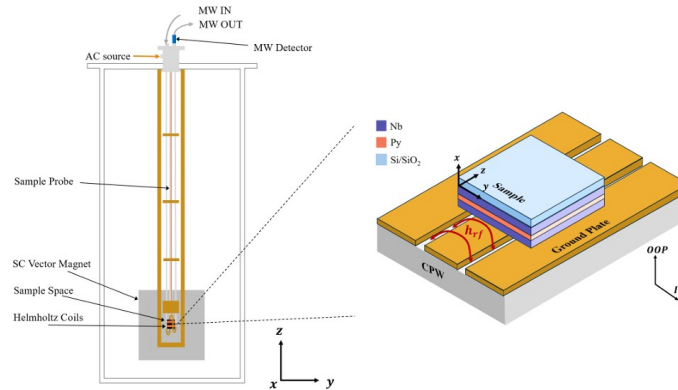


Figure 2.4: Schematic diagram of the cryostat and probe assembly.

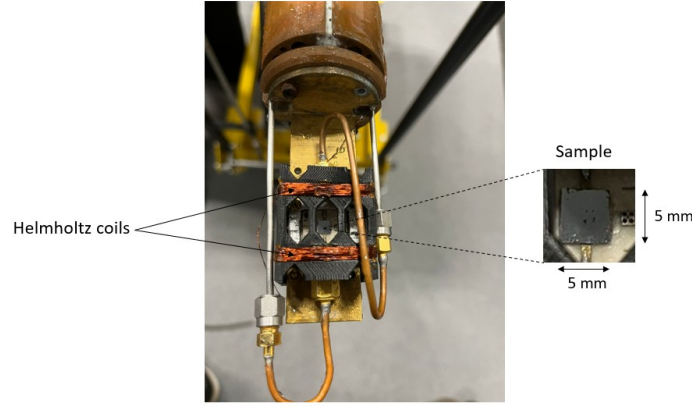


Figure 2.5: Picture of the sample and probe assembly.

Fig. 2.6 shows a schematic of the complete setup used to conduct the FMR measurements. A signal is sent from a microwave source to the CPW through a coaxial cable, the current flowing in the CPW creates an oscillating magnetic field at the same microwave frequency as the input signal. According to Biot–Savart law, this field lies along the y axis and is approximately uniform across the sample. Together with the static magnetic field produced by the superconducting coil of the cryostat, which is constant in time, approximately uniform along the sample, and oriented along the z axis, this oscillating field can, in principle, induce ferromagnetic resonance. However, this effect is difficult to detect directly due to noise; therefore, a universal source, a pair of Helmholtz coils, and a lock-in amplifier are introduced to improve signal detection. Part of the energy carried by the signal is absorbed by the sample through FMR. The resulting transmitted signal is then sent to a diode, a composite component that converts the AC signal into a DC signal with an amplitude proportional to the average power of the input signal. The universal source sends a signal at 321 Hz to the two Helmholtz coils, generating a magnetic field whose frequency is much lower than the MW frequency of the input signal. This field can therefore be considered constant in time and added to the DC magnetic field generated by the electromagnet, the lock-in amplifier then registers the variation in the diode voltage δV_{diode} caused by a small variation δH_{DC} of the DC magnetic fields and returns a signal proportional to the ratio $\frac{\delta V_{diode}}{\delta H_{DC}}$ which is approximately equal to the derivative of V_{diode} with respect to H_{DC} .

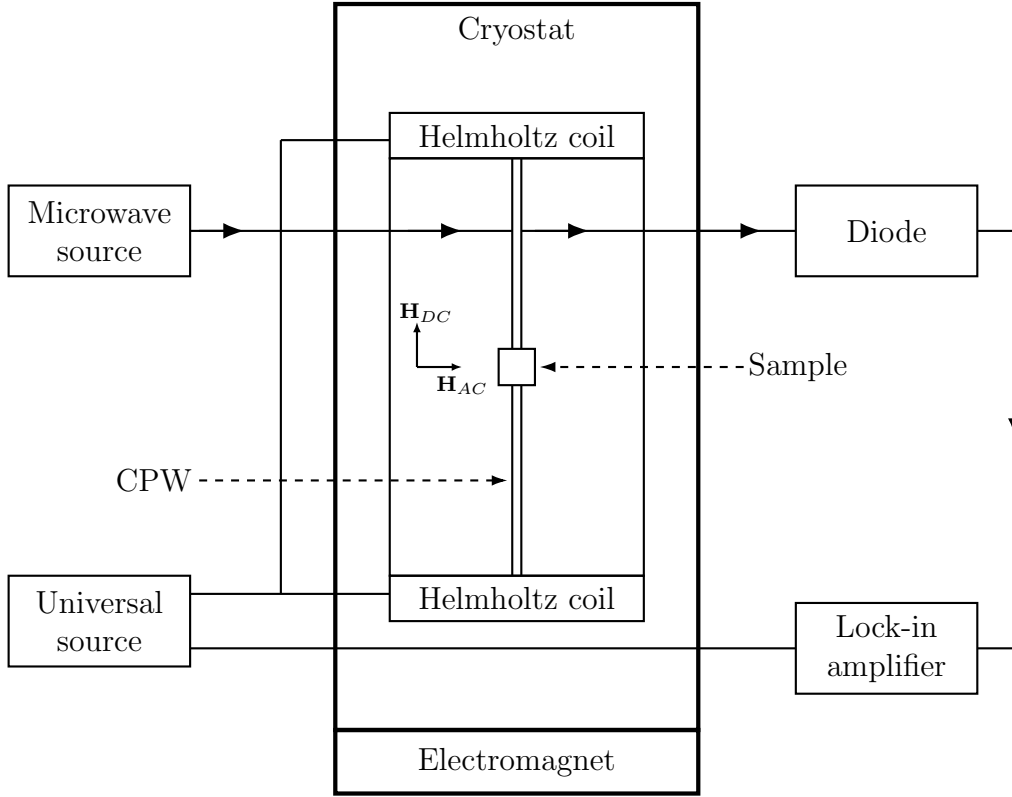


Figure 2.6: Schematic illustration of the setup used in the experiment.

2.3 FMR Measurements

Ferromagnetic resonance can be conducted by either sweeping frequency or magnetic field. In this latter case, the FMR absorption spectrum is detected by changing the external magnetic field at fixed frequencies to find the resonance field. As explained in the previous section, the resulting data is in the form of the derivative of the Lorentzian

$$V = S \frac{\Delta H^2 (H - H_{res})}{(\Delta H^2 + (H - H_{res})^2)^2} + A \frac{\Delta H (H - H_{res})^2 - \Delta H^3}{(\Delta H (H - H_{res})^2)^2} + aH + d \quad (2.1)$$

which is analysed using the symmetric and antisymmetric lorentzian functions with respect to H_{res} . Another effect that is observed is that the data present a linear noise that is added to the lorentzian derivative function, therefore the profile of the data is expected to look like Fig. 2.7. The origin of the linear noise is still not known. A possible explanation for the linear noise is an impedance mismatch in the connections between the microwave generator and the CPW and between the CPW and the diode.

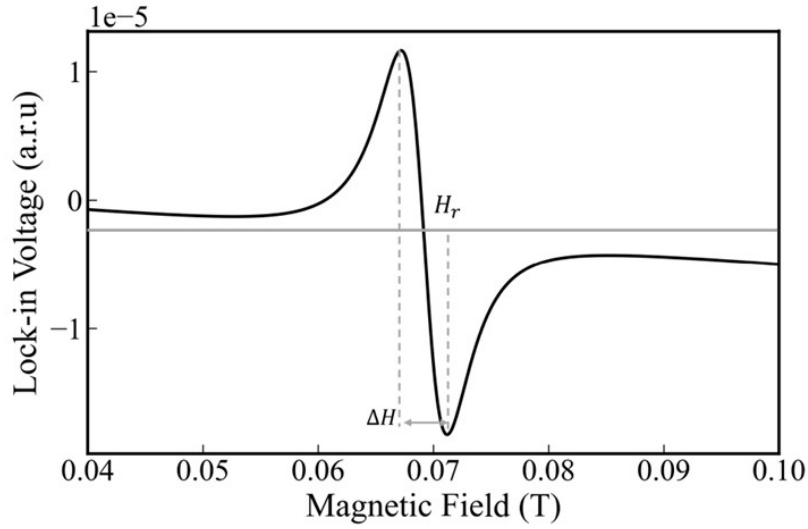


Figure 2.7: Expected behaviour of the signal

From this, the resonant field and linewidths can be extracted from each scan at each frequency and temperature by doing a fit on the data with a lorentzian derivative combined with a linear background. The values of resonant field and resonant frequency can be analyzed to form the Kittel relationship, allowing for exploration of changes in the resonant frequency-resonant field relationship. Apart from the bilayer sample, which is expected to follow the standard Kittel equation 1.31, the other samples are expected to exhibit anisotropy consistent with the modified Kittel equation 1.66.

2.4 Summary

This chapter provided an overview of the instruments used in the study. By using the setup described in Section 2.2 to perform the field-swept FMR measurements outlined in Section 2.3 on the samples described in Section 2.1, the results presented in the next chapter were obtained.

Chapter 3

Data analysis and Results

This chapter presents the data obtained from the FMR measurements and the results of their analysis. The goals of the experiment were to use the setup described in the previous chapter to confirm the results of Refs. [1], [2] and [3] (section 3.1) and to test the theory from Ref. [9] (section 3.2) by performing FMR measurements on the samples in tables 2.1 and 2.2. Section 3.3 summarizes the experimental findings of this study.

3.1 Reproducing Literature Results

The first step of the investigation was to reproduce the results reported in the literature. Specifically, it is known that introducing two superconducting thin film layers around a ferromagnetic thin film generates a shift in the resonant field H_{res} , which therefore follows the modified Kittel equation (1.66) instead of the standard one (1.31). This shift appears only below a certain critical temperature, and both superconducting layers are required for the effect to occur. The first two conditions were verified using the control sample, while the last one was confirmed with the bilayer sample.

3.1.1 Control Sample

For different values of the temperature T and frequency f , the lock-in voltage was measured as a function of the applied DC magnetic field. All measurements were performed from 10 K down to 1.5 K over a frequency range of 6 to 15 GHz, using field-modulation FMR. As explained in the previous chapter, the signal detected by the lock-in amplifier is proportional to the derivative with respect to the magnetic field H of the signal from the CPW. Therefore, the data were fitted using Eq. (2.1). From this fit, the resonant field can be

identified as the point where the function, excluding the linear background, crosses zero. Fig. 3.1 shows a representative plot of the measured data.

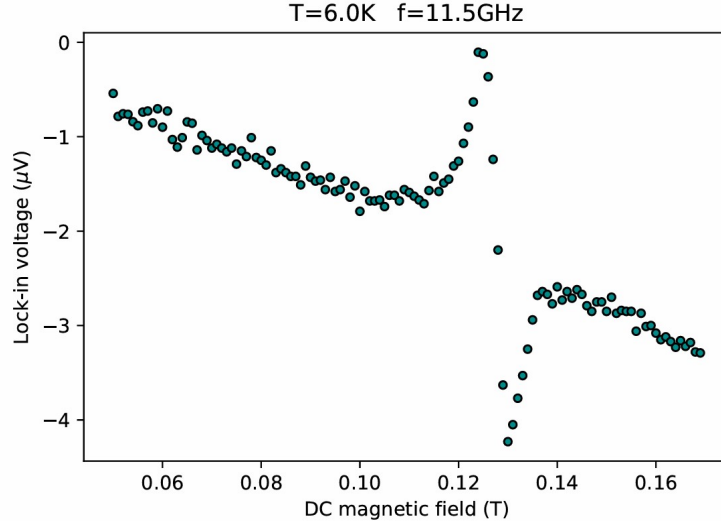


Figure 3.1: Representative experimental data for the lock-in voltage.

As expected, the plot follows the qualitative behaviour shown in Fig. 2.7. To fit the data, the linear background was first removed. This was done by performing a linear fit over the entire dataset. Fig. 3.2 shows the data from Fig. 3.1 together with their fitted linear component. The linear component was then subtracted from the data and a four parameter fit was made on the resulting dataset. The initial guess values for the parameters H_{res} , ΔH , S , A were calculated using the following formulas:

$$H_{guess}^{res} = \frac{H_{max} + H_{min}}{2} \quad (3.1)$$

$$\Delta H_{guess} = 0.1(H_{max} - H_{min}) \quad (3.2)$$

$$S_{guess} = 0.7(V_{max} - V_{min}) \quad (3.3)$$

$$A_{guess} = 0.3(V_{max} - V_{min}) \quad (3.4)$$

where V_{max} and V_{min} are the maximum and minimum value of the lock-in voltage in the data, while H_{max} and H_{min} are the corresponding values of the DC magnetic field. Fig. 3.3 shows the same dataset together with the final fitted curve.

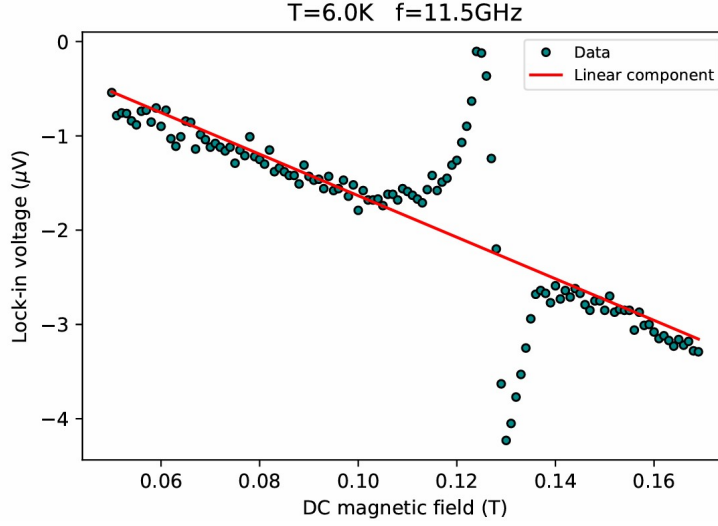


Figure 3.2: Linear component extrapolated from a fit on the entire dataset.

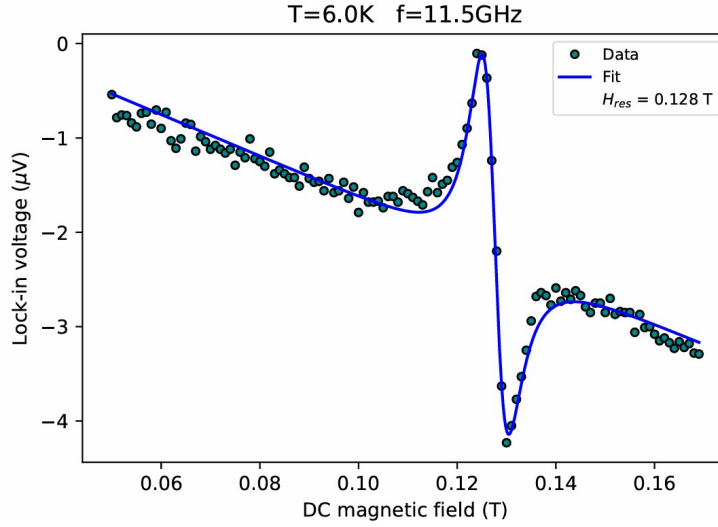


Figure 3.3: Fit curve obtained from the guess (3.1) - (3.4).

For each temperature the values of resonant frequency and field were fitted with the modified Kittel's Eq. 2.7 to find the anisotropy field H_a . Fig. 3.4 shows an example of a dataset together with its fitted curve. Fig. 3.5 all the fitted curves at different temperature for a comparison.

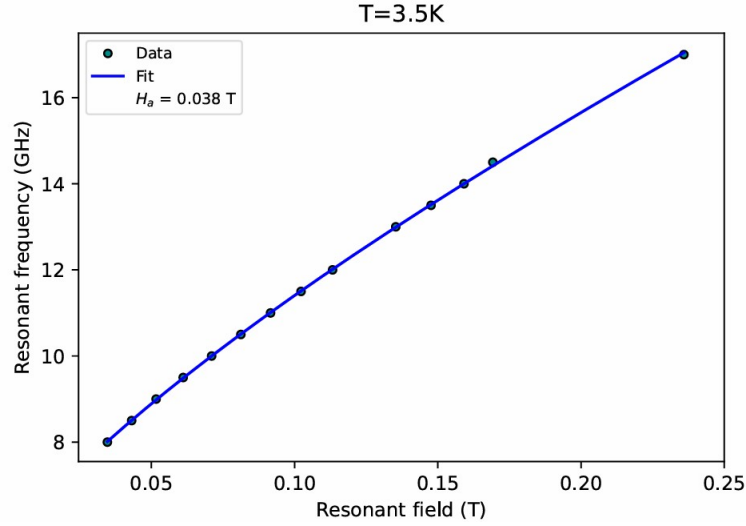


Figure 3.4: Experimental data for resonant field and resonant frequency at $T = 3.5\text{ K}$.

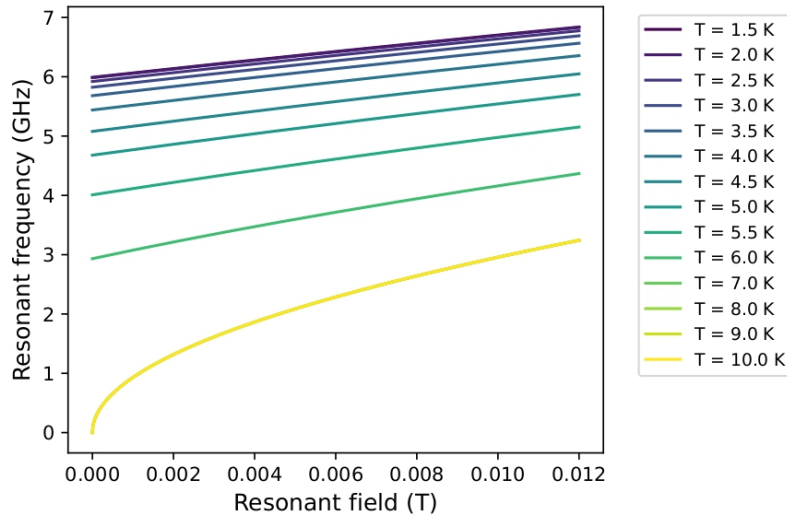


Figure 3.5: Fitted curves of the Kittel equation for different temperatures.

As expected, the curves above the critical temperature coincide, confirming that no change in the anisotropy occurs in this regime. In contrast, the curves below the critical temperature display an offset that increases as the temperature decreases. Finally, the values of the anisotropy field H_a were plotted as a function of the temperature T to verify that the shift in H_{res}

appears only below the critical temperature T_c . The resulting plot is shown in Fig. 3.6.

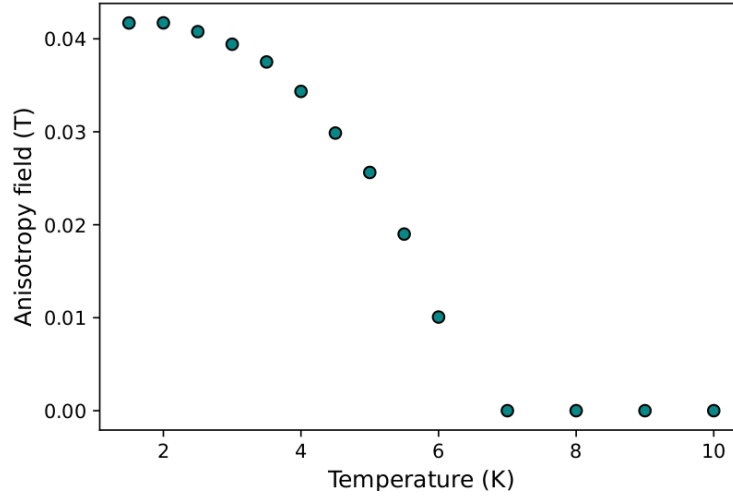


Figure 3.6: Anisotropy field as a function of temperature.

As expected, H_a has a constant value of zero below T_c . The observed critical temperature is lower than the theoretical critical temperature of niobium (9 K), this can be attributed to a proximity effect.

The data for the anisotropy field with $H_a > 0$ were then fitted with the power law showed in section 1.3

$$H_a(T) = H_0 \left(1 - \left(\frac{T}{T_c} \right)^p \right) \quad (3.5)$$

and the following values for the critical temperature T_C and for the exponent p were obtained

$$T_c = 6.5 \pm 0.1 \text{ K} \quad (3.6)$$

$$p = 3.5 \pm 0.1 \quad (3.7)$$

Fig. 3.7 shows the comparison between the data and the fit curve.

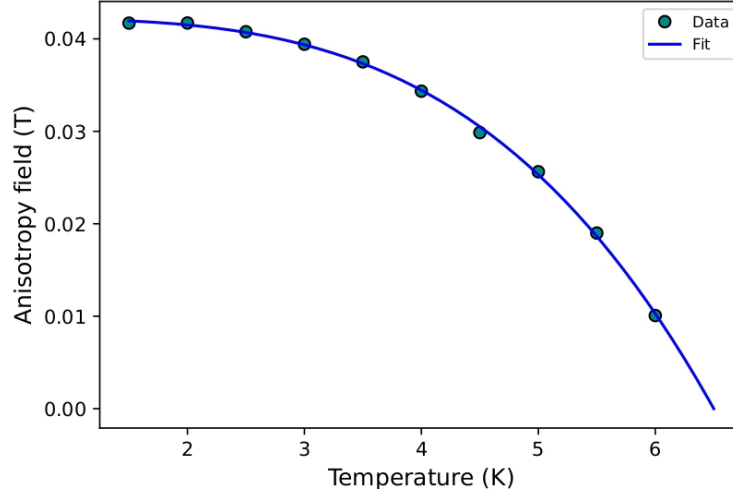


Figure 3.7: Values of the anisotropy field with the fit curve obtained from the power law (3.5) below T_c .

3.1.2 Bilayer

The same procedure used for the control sample was used for the bilayer sample. Fig. 3.8 shows the values of the anisotropy field for different temperatures both above and below the critical temperature.

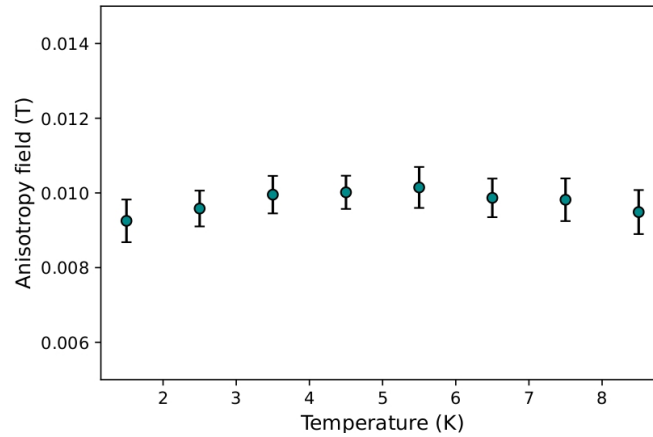


Figure 3.8: Anisotropy field for different temperatures in the asymmetric sample.

In this case, no change in the anisotropy field is observed below the critical

temperature within the experimental errors. This indicates that both superconducting layers are required in order to produce the change in anisotropy. Moreover, the anisotropy field remains greater than zero even below the critical temperature, this is due to shape anisotropy and impurities in the sample.

3.2 Additional experiments

The next step of the investigation was to test the theory presented in Ref. [9] by conducting FMR measurements on a thin film multilayer containing two additional metallic (Cu) layers, as well as on a trilayer at different microwave power levels. The aim is to determine whether the effect exhibits any dependence on the microwave power, which is expected for a dynamical mechanism such as the one proposed in Ref. [9], and whether the introduction of metallic layers (through which electric currents can flow) leads to any modification of the observed behavior.

3.2.1 Copper Thickness

The same procedure described in the previous section was used to extract the values of the resonant field for samples with additional copper layers of thicknesses ranging from 2 nm to 100 nm (table 2.2). For the sample with a 200 nm Cu layer, it was not possible to obtain reliable data because, in this case, the ferromagnetic layer was too far from the CPW and the resulting microwave magnetic field was too weak to be detected. The anisotropy field values for the different samples below the critical temperature are shown in Fig. 3.9.

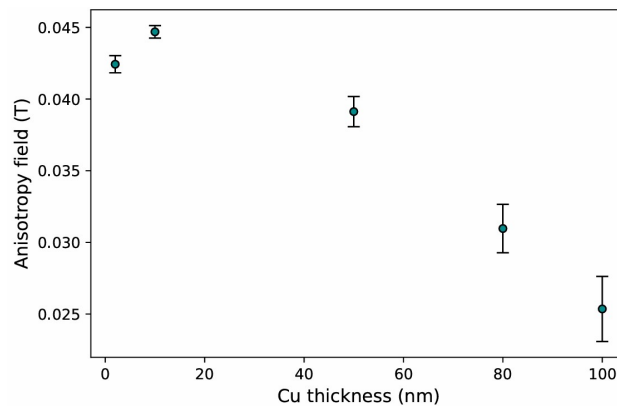


Figure 3.9: Anisotropy field at $T = 2.0K$ for different thicknesses of the Cu layers.

Although the plot does not contain enough data points to draw a definitive conclusion about the behavior of the anisotropy field as a function of the copper thickness, it clearly shows that the effect does not disappear upon introducing a conducting layer, as it does when an insulating layer is inserted. This observation suggests that the effect is linked to currents flowing through the entire system.

3.2.2 Microwave Power

The values of the resonant field were extracted from the Kittel fits at different microwave powers for a trilayer sample (table 2.1). The corresponding resonant field for each value of the microwave power are shown for in Fig. 3.10. A variation in the resonant field is observed; however, this is most likely due

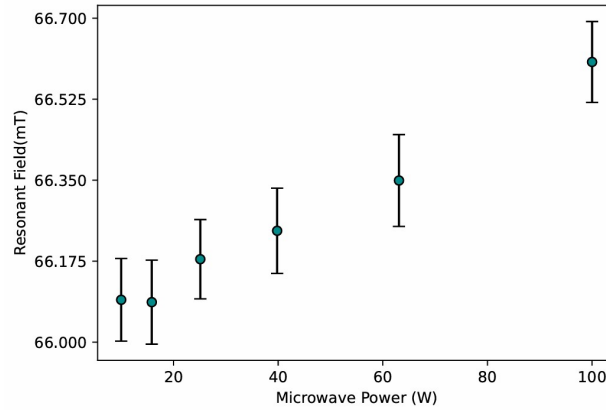


Figure 3.10: Anisotropy field at $T = 2.0K$ for different values of the microwave power.

to heating of the system. The microwave signal induces eddy currents in the trilayer, which raise its temperature and thereby reduce the anisotropy field, resulting in a small shift of the resonant field. It can therefore be concluded that the effect is independent of the microwave power.

3.3 Results

The first part of the study confirmed the core results reported in earlier works: when the ferromagnetic layer is coupled with two superconducting layers, the resonant field extracted from ferromagnetic resonance (FMR) measurements deviates from the standard Kittel relation (1.31) and instead follows a modified form (1.66) containing an additional anisotropy field. This additional term appears only below the superconducting critical temperature and increases as the temperature decreases, demonstrating a clear connection between the effect and the superconducting phase. Furthermore, the bilayer measurements showed no detectable change in the resonant field across the superconducting transition, confirming that both superconducting layers are required for the effect to occur. These outcomes reproduce the essential features found in the literature (Refs. [1], [2], [3]).

In the second part of the work, other experiments were performed to probe the origin of the additional anisotropy field. Measurements on multilayers containing copper spacers of varying thicknesses showed that the effect persists even when metallic layers are inserted between the superconducting and ferromagnetic layers. In contrast, previous studies have shown that inserting an insulating layer suppresses the effect entirely. This comparison suggests that the phenomenon is associated with electrical currents flowing through the multilayer system, rather than arising from a purely local interfacial mechanism. Additional measurements were carried out at different microwave power levels to determine whether the shift might originate from a dynamical mechanism, as suggested in Ref. [9]. The experimental results showed small variations in the resonant field with microwave power that are consistent with thermal effects due to microwave-induced heating. No significant dependence on the microwave power was observed, indicating that the effect does not originate from the dynamical effect within the accessible power range.

From the experiments conducted in this investigation it can be concluded that the model presented in Ref. [9] does not describe all the features of the phenomenon in exam.

Conclusions

This thesis presented an experimental investigation of the superconductivity-induced shift in the ferromagnetic resonant field observed in superconductor–ferromagnet–superconductor (S/F/S) trilayers. After explaining the main features of ferromagnetic and superconducting materials and the proximity effects in interfaces between these materials, the second chapter presents the main results known in literature about these phenomenon. The third chapter then presented the experimental setup involved in the measurements and the PVD technique used to grow the samples studied. Finally the fourth chapter presents the data obtained from the measurements and the results of the study.

The results obtained so far showed that this effect enhances the anisotropy in the ferromagnet plane. However, in order to use this effect for switching applications, it is necessary to change the direction of the magnetization in a direction perpendicular to the plane of the ferromagnetic layer. Therefore the next step is to focus on ferromagnets with some perpendicular magnetic anisotropy (PMA).

Other models that aim to explain the change in anisotropy in SFS heterostructures could also be tested. In Ref. [11], an alternative explanation is presented. In this model, the precessing magnetization of the ferromagnetic layer induces, according to London theory, an electric field in the superconducting layers, which in turn generates a current. This current produces an additional magnetic field that shifts the resonance field and frequency. The model correctly predicts the necessity of having both superconducting layers. In the same paper, FMR measurements are performed on different samples that are consistent with the predictions of the model. A possible direction for future work would be to design experiments specifically aimed at testing the predictions of this theory.

Finally, Another possible development can be to replicate these same FMR measurements on an inverted FSF structure to see if the central superconducting layer creates a coupling between the two external ferromagnetic layers and produces an anisotropy field in both layers.

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Scannerizzare il QR-code per la versione semplificata dei ringraziamenti.



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